

Continuous random variables

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Subsection 1

Introduction

Why continuous random variables?

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- Suppose that $|\mathcal{X}|$ is growing very large and unwieldy for calculations

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 - ★ Data available for 45000+ meteorites
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 - ★ n grows very large
 - ★ p is a constant
 - ★ can happen in Bernoulli trials

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 - ★ can happen in Bernoulli trials
- Is there a way to simplify the descriptions of these random-like phenomena?
 - ▶ Yes! But we need to give up something. . .

Meteorite data

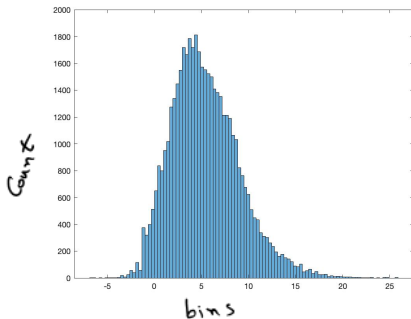
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 - ▶ Range: from $[0.01, 600000000]$ to $[-6.6, 25.8]$
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- Main idea: Move from individual values to intervals of values
 - ▶ Divide $[-6.6, 25.8]$ into ≈ 100 intervals
 - ★ $[-6.6, -6.3], [-6.3, -6], \dots, [25.5, 25.8]$
 - ▶ Count the number of values falling inside each interval

Meteorite data

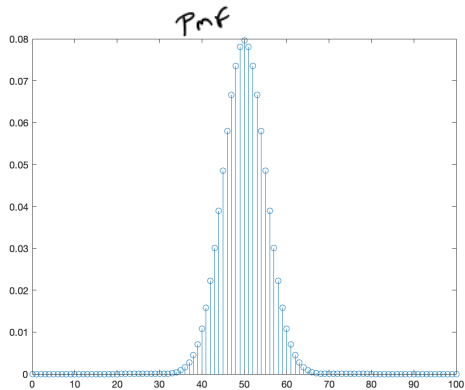
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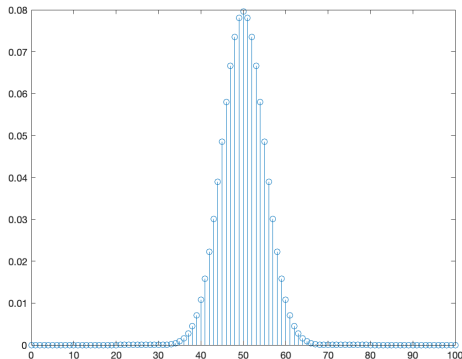
Histogram of log of weights

- Describe shape of the histogram instead of individual values
- What do we give up? Precision is reduced

Bernoulli trials, $n = 100$, $p = 0.5$



Bernoulli trials, $n = 100$, $p = 0.5$



- Calculations with the PMF is not very easy even in this case
- Working with intervals and histograms can be much simpler

Subsection 2

~~Cumulative~~ distribution function

CDF of a random variable

Definition (CDF of a random variable)

The Cumulative Distribution Function (CDF) of a random variable X , denoted $F_X(x)$, is a function from $\mathbb{R}$ to $[0, 1]$, defined as

$$F_X(x) = P(X \leq x).$$

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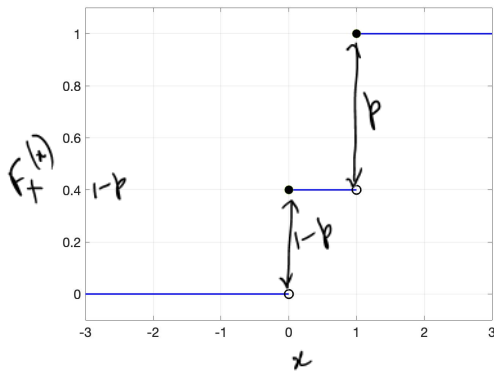
$$F_X(x) = P(X \leq x).$$

Properties

- $F_X(b) - F_X(a) = P(a < X \leq b)$
- F_X : non-decreasing function taking non-negative values
- As $x \rightarrow -\infty$, F_X goes to 0.
- As $x \rightarrow \infty$, F_X goes to 1.

Example: Bernoulli random variable

$$X \sim \left\{ \underline{\underline{0}}^{1-p}, \underline{\underline{1}}^p \right\}$$



$$F_X(x) = \begin{cases} 0 & x < 0 \quad \checkmark \\ 1-p & 0 \leq x < 1 \quad \checkmark \\ 1 & 1 \leq x \quad \checkmark \end{cases}$$

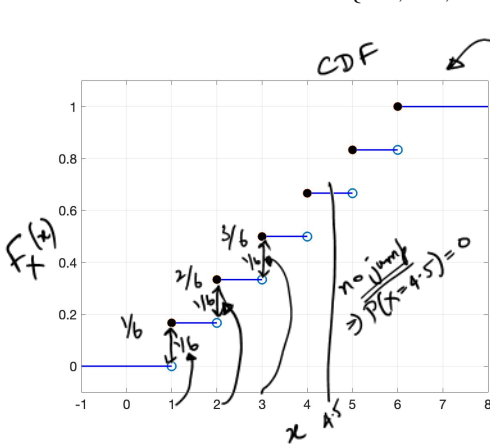
$$F_X(-1) = \mathcal{P}(X \leq -1) = 0$$

$$F_X(x) = \mathcal{P}(X \leq \underline{\underline{0.1}}) = \mathcal{P}(X=0) = 1-p$$

$x = 0.1$ any $x: 0 \leq x \leq 1$

Example: Throw a die

$$X \sim \left\{ \overset{1/6}{1}, \overset{1/6}{2}, \overset{1/6}{3}, \overset{1/6}{4}, \overset{1/6}{5}, \overset{1/6}{6} \right\} \text{ PMF}$$



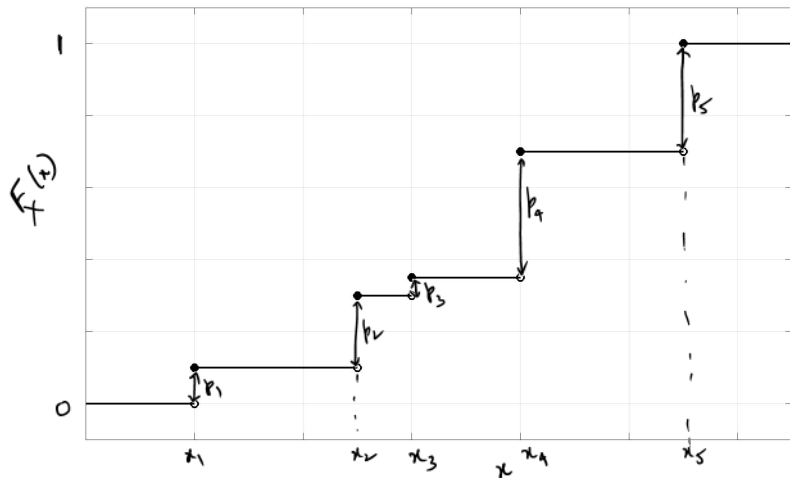
$$F_X(x) = \begin{cases} 0 & x < 1 \\ 1/6 & 1 \leq x < 2 \\ 2/6 & 2 \leq x < 3 \\ 3/6 & 3 \leq x < 4 \\ 4/6 & 4 \leq x < 5 \\ 5/6 & 5 \leq x < 6 \\ 1 & 6 \leq x \end{cases}$$

CDF of a discrete random variable

$$X \sim \{x_1^{p_1}, x_2^{p_2}, x_3^{p_3}, x_4^{p_4}, x_5^{p_5}\}$$

CDF of a discrete random variable

$$X \sim \{X_1, X_2, X_3, X_4, X_5\}$$



Computing probability of intervals using CDF

$$X \sim \text{Uniform}\{1, 2, \dots, 100\}$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ k/100 & k \leq x < k+1, k = 1, 2, \dots, 99 \\ 1 & x \geq 100 \end{cases}$$

Handwritten notes:

- $x < 0$ is underlined.
- An arrow points from the middle case to the interval $1 \leq x < 2$ with $\frac{1}{100}$ written above it.
- Below that, $2 \leq x < 3$ is written with $\frac{2}{100}$ written above it.
- Vertical ellipsis follows.

Computing probability of intervals using CDF

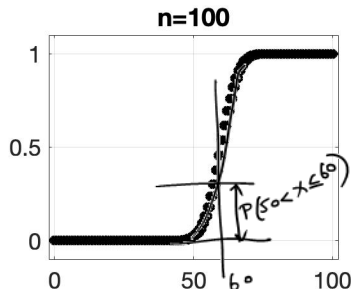
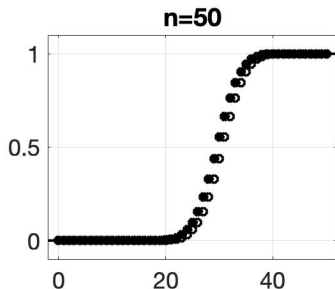
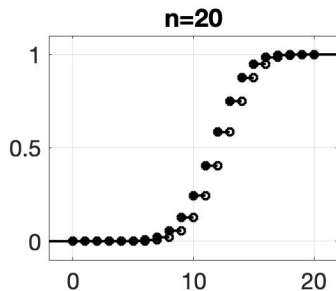
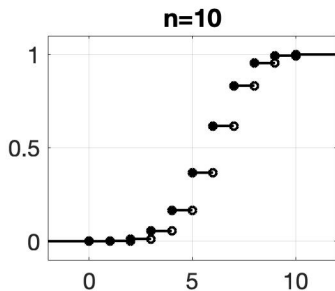
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- $P(3 < X \leq 10) = F_X^{10/100}(10) - F_X^{3/100}(3) = 7/100$
- $P(3.2 < X \leq 10.6) = F_X^{10.6/100}(10.6) - F_X^{3.2/100}(3.2) = 7/100$
- $P(X \leq 17) = F_X(17) = 17/100$
- $P(X \leq 17.3) = F_X(17.3) = 17/100$
- $P(\underline{X > 87}) = 1 - \underline{F_X(87)} = 13/100$
- $P(X > 87.4) = 1 - \underline{F_X(87.4)} = 13/100$

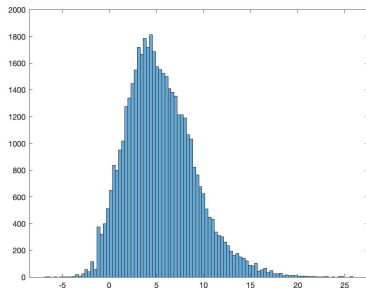
$$\begin{aligned} P(X > 87) &= 1 - P(X \leq 87) \\ &= 1 - P(X \leq 87) \\ &= 1 - F_X(87) \end{aligned}$$

Large alphabet: Binomial($n, 0.6$)

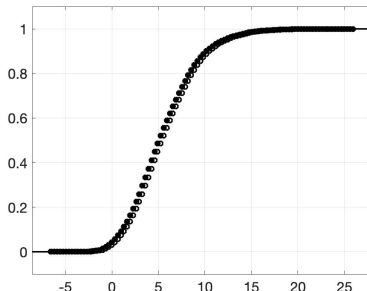


Large alphabet: Meteorite data

Histogram of weights



CDF of weight distribution



Cumulative Distribution Functions

Definition (CDF)

A function $F : \mathbb{R} \rightarrow [0, 1]$ is said to be a Cumulative Distribution Function (CDF) if

- 1 F is a non-decreasing function taking values between 0 and 1.
- 2 As $x \rightarrow -\infty$, F_x goes to 0.
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- 4 Technical: F is continuous from the right.

Cumulative Distribution Functions

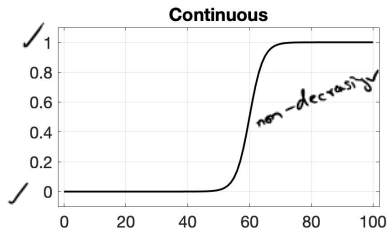
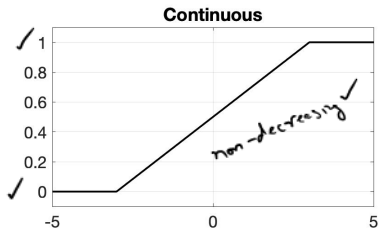
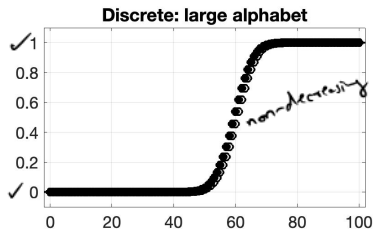
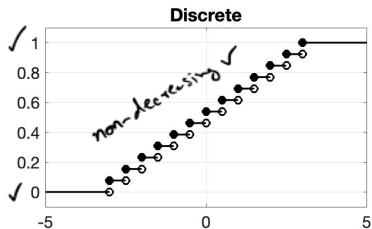
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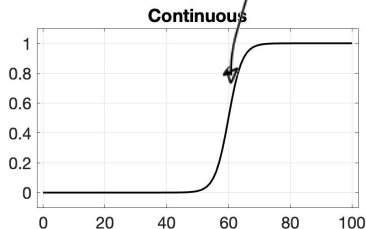
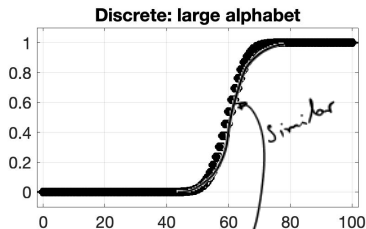
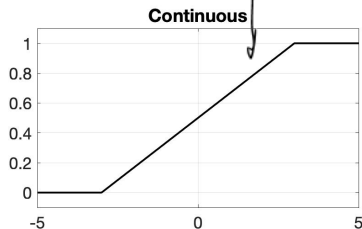
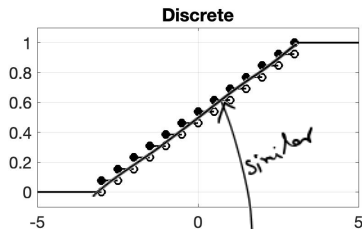
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- Definition motivated by CDF of a random variable defined earlier
- A general CDF need not be like a CDF of a discrete random variable
 - ▶ No need for a step-like structure
 - ▶ Can be smooth and continuous

Examples of valid CDFs



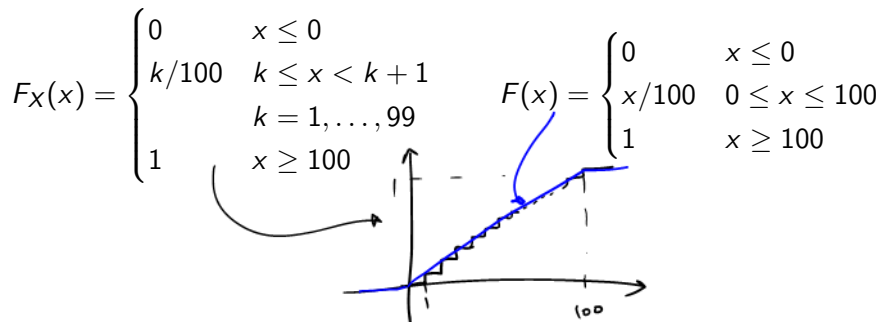
Examples of valid CDFs



- Continuous CDFs appear to be close approximations to CDFs of discrete random variables, particularly when alphabet grows.

Probability of intervals using continuous CDF

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Simpler continuous CDF approximates prob of intervals!

Binomial using continuous CDF

$$X \sim \text{Binomial}(100, 0.6)$$

$$F_X(k) = \sum_{j=0}^k \binom{100}{j} (0.6)^j (0.4)^{100-j}$$

$k = 0, 1, \dots, 100$

$$F(x) = \frac{1}{1 + \exp\left(\frac{-1.65451(x-60)}{\sqrt{24}}\right)}$$

Handwritten annotations: n above the 100 in the binomial coefficient, and $np(1-p)$ below the $\sqrt{24}$ in the denominator of the normal approximation.

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- $P(40 < X \leq 50) = 0.0271$, $F(50) - F(40) = 0.0318$
- $P(50 < X \leq 60) = 0.5108$, $F(60) - F(50) = 0.4670$
- $P(60 < X \leq 70) = 0.4473$, $F(70) - F(60) = 0.4670$
- $P(70 < X \leq 80) = 0.0148$, $F(80) - F(70) = 0.0318$

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Better approximations are possible, particularly as n grows!

Subsection 3

General random variables and Continuous random variables

CDFs and random variables

Theorem (Random variable with CDF $F(x)$)

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Properties

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- $P(a < X \leq b) = F(b) - F(a)$
- If $F(x)$ rises from F_1 to F_2 at x_1 , $P(X = x_1) = F_2 - F_1$

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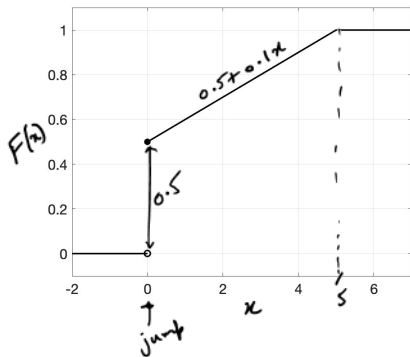
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Properties

- $P(a < X \leq b) = F(b) - F(a)$
- If $F(x)$ rises from F_1 to F_2 at x_1 , $P(X = x_1) = F_2 - F_1$
- If $F(x)$ is continuous at x_0 , $P(X = x_0) = 0$ (non-intuitive!)

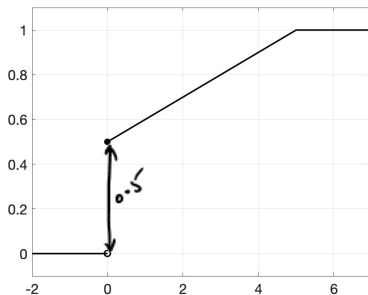
Example: Random variable X with CDF $F(x)$

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.5 + 0.1x & 0 \leq x \leq 5 \\ 1 & x > 5 \end{cases}$$



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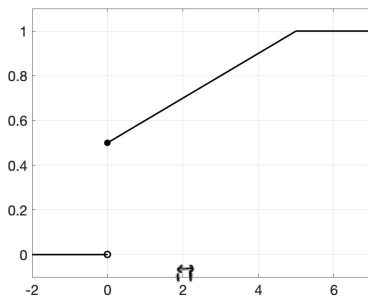
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- $P(X = 0) = 0.5$

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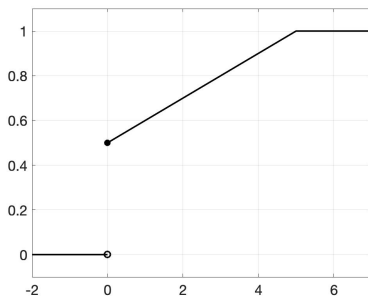
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- $P(X = 0) = 0.5$
- $P(1.99 < X \leq 2.01) = F(2.01) - F(1.99) = 0.002$

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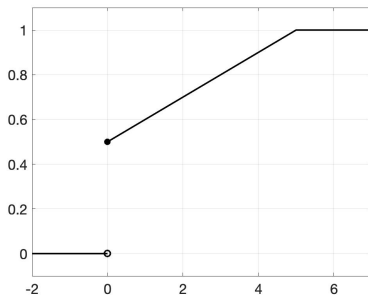
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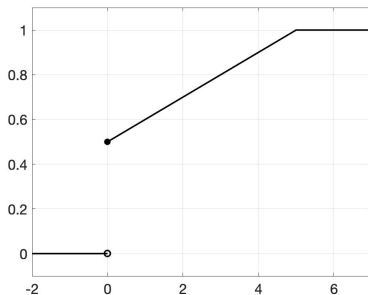
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 - ▶ Value with finite precision taken with positive probability
- $P(1.9999999 < X \leq 2.0000001) = 0.00000002$

Example: Random variable X with CDF $F(x)$

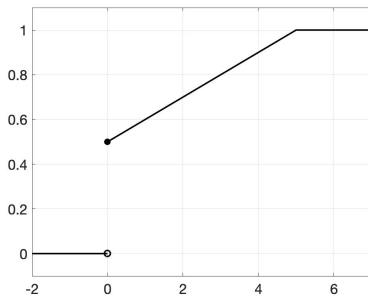
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 - ▶ As precision increases, probability decreases

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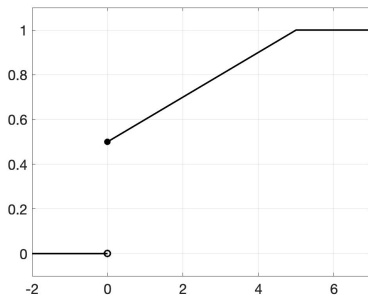
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 - ▶ Value with finite precision taken with positive probability
- $P(1.9999999 < X \leq 2.0000001) = 0.00000002$
 - ▶ As precision increases, probability decreases
- $P(X = 2.00000 \dots) = 0$

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 - ▶ Value with finite precision taken with positive probability
- $P(1.9999999 < X \leq 2.0000001) = 0.00000002$
 - ▶ As precision increases, probability decreases
- $P(X = 2.00000 \dots) = 0$
 - ▶ Values with infinite precision cannot be taken, when $F(x)$ is continuous at that point

Continuous random variable

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- CDF has no jumps or steps
- So, $P(X = x) = 0$ for all x
- Probability of X falling in an interval will be nonzero

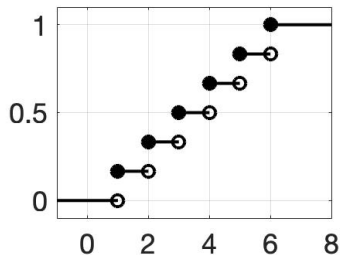
$$P(a < X \leq b) = F(b) - F(a)$$

- Since $P(X = a) = 0$ and $P(X = b) = 0$, we have

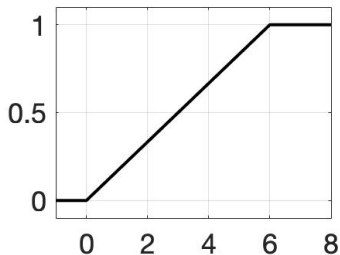
$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$$

Examples

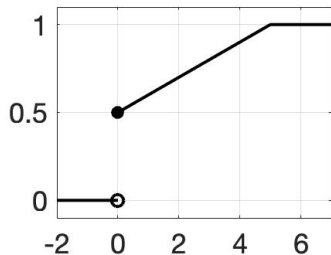
Discrete



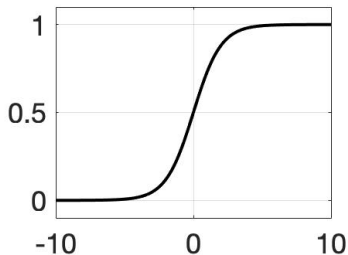
Continuous



Mixture



Continuous



Some scenarios for continuous models

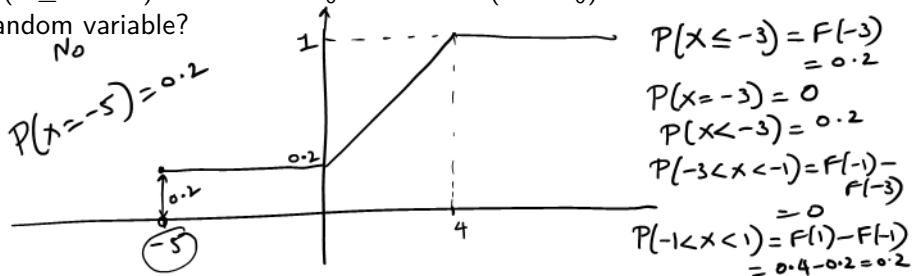
- Throw a dart onto a circular board - distance of the point of impact from the center of the board.
- Weight of a meteorite hitting earth
- Weight of a human being, height of a human being
- Speed of a delivery in cricket
- Price of a stock
- Many discrete random variables are well-approximated by continuous random variables that are much simpler to describe

Problem

Consider a random variable X with CDF

$$F_X(x) = \begin{cases} 0 & x < -5 \\ 0.2 & -5 \leq x < 0 \\ 0.2 + 0.2x & 0 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

Find $P(X < -3)$, $P(-3 < X < -1)$, $P(-1 < X < 1)$, $P(X \leq -3)$, $P(X \geq 3)$, $P(0 \leq X < 3)$. Is there an x_0 for which $P(X = x_0) > 0$? Is X a continuous random variable?



Working

$$\begin{aligned}P(X > -2) &= 1 - P(X \leq -2) = 1 - F(-2) \\&= 1 - 0.2 = 0.8\end{aligned}$$

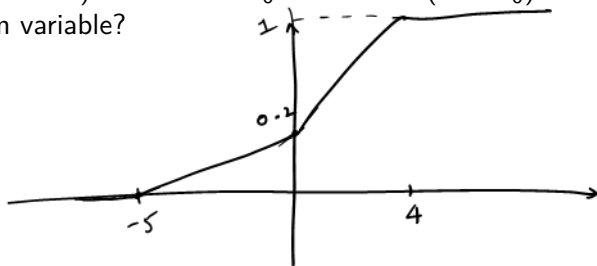
$$P(X \geq 3) = P(X > 3) = 1 - P(X \leq 3) = 1 - F(3) = 1 - 0.8 = 0.2$$

Problem

Consider a random variable X with CDF

$$F_X(x) = \begin{cases} 0 & x < -5 \\ 0.04x + 0.2 & -5 \leq x < 0 \\ 0.2 + 0.2x & 0 \leq x < 4 \\ 1 & x \geq 4. \end{cases}$$

Find $P(X < -3)$, $P(-3 < X < -1)$, $P(-1 < X < 1)$, $P(X \leq -3)$, $P(0 \leq X < 3)$. Is there an x_0 for which $P(X = x_0) > 0$? Is X a continuous random variable? Yes



Working

Subsection 4

Probability density function and common continuous distributions

Refresher on integration

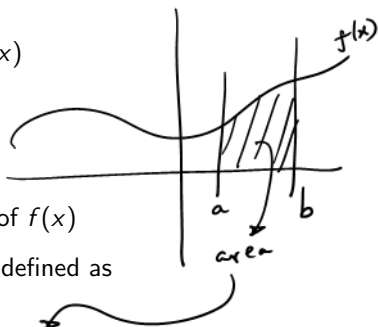
- Indefinite integral of a function $f(x)$

- ▶ A function $F(x)$ such that $\frac{dF(x)}{dx} = f(x)$
- ▶ Denoted as $F(x) = \int f(x) dx$

- Definite integral of a function $f(x)$

- ▶ Suppose $F(x)$ is the indefinite integral of $f(x)$
- ▶ Definite integral of $f(x)$ from a to b is defined as

$$\int_a^b f(x) dx = F(b) - F(a)$$



- ▶ Definite integral equals the area under the curve $f(x)$ from a to b

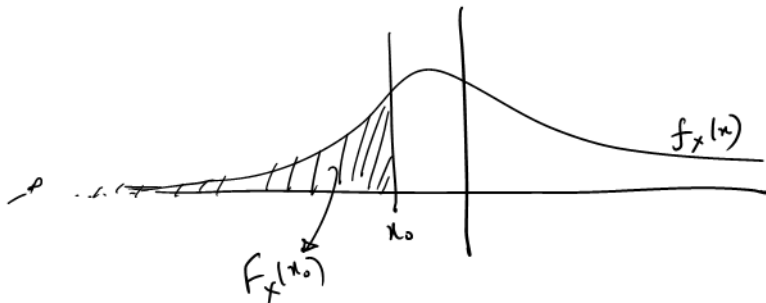
Tables of integrals: https://en.wikipedia.org/wiki/Lists_of_integrals

Probability density function (PDF)

Definition (PDF)

A continuous random variable X with CDF $F_X(x)$ is said to have a PDF $f_X(x)$ if, for all x_0 ,

$$F_X(x_0) = \int_{-\infty}^{x_0} f_X(x) dx.$$



Probability density function (PDF)

Definition (PDF)

A continuous random variable X with CDF $F_X(x)$ is said to have a PDF $f_X(x)$ if, for all x_0 ,

$$F_X(x_0) = \int_{-\infty}^{x_0} f_X(x) dx.$$

- CDF is the integral of the PDF
 - ▶ Derivative of the CDF (wherever it exists) is usually taken as the PDF

PDF: $f_X(x)$ Anti-derivative: $F_X(x)$

$$\int_{-\infty}^{x_0} f_X(x) dx = F_X(x_0) - \underbrace{F_X(-\infty)}_{=0}$$

Probability density function (PDF)

Definition (PDF)

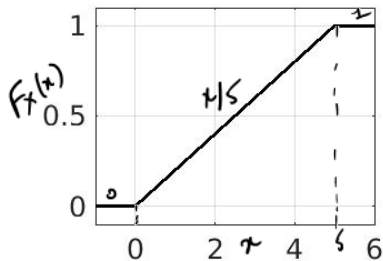
A continuous random variable X with CDF $F_X(x)$ is said to have a PDF $f_X(x)$ if, for all x_0 ,

$$F_X(x_0) = \int_{-\infty}^{x_0} f_X(x) dx.$$

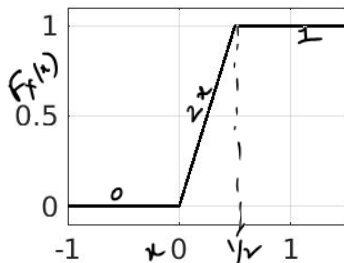
- CDF is the integral of the PDF
 - ▶ Derivative of the CDF (wherever it exists) is usually taken as the PDF
- Why PDF?
 - ▶ Value of PDF around $f_X(x_0)$ is related to X taking a value around x_0
 - ★ Higher the PDF, higher the chance that X lies there
 - ★ Contrast with value of CDF at x_0 , $F_X(x_0)$
 - ★ PDF is much easier in probability computations

Examples: Uniform distribution

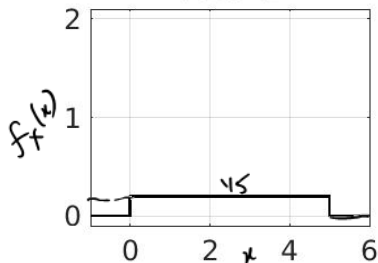
CDF 1



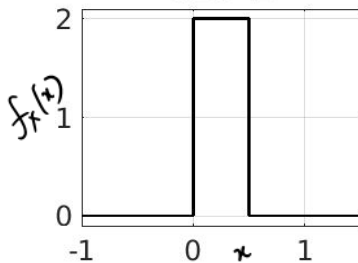
CDF 2



PDF 1

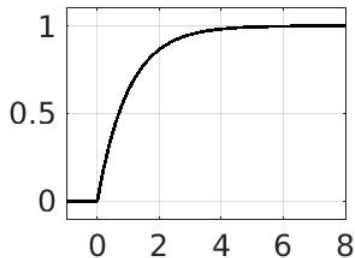


PDF 2

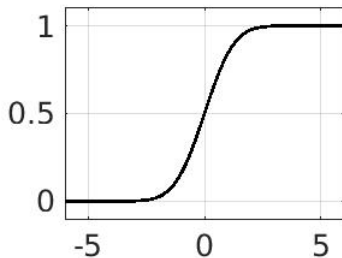


Examples: Exponential and normal distribution

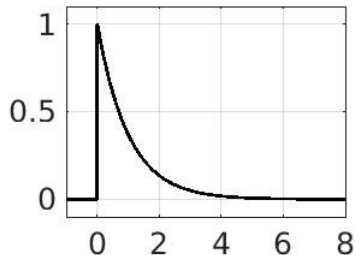
CDF 1



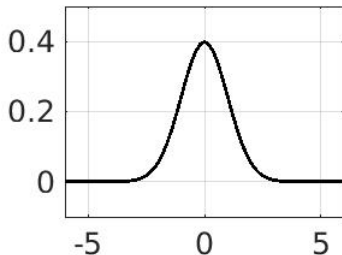
CDF 2



PDF 1



PDF 2



Properties of PDF

Definition (Density function)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be a density function if

- 1 $f(x) \geq 0$
- 2 $\int_{-\infty}^{\infty} f(x)dx = 1$
- 3 $f(x)$ is piecewise continuous

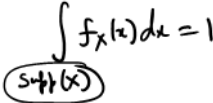
Properties of PDF

Definition (Density function)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be a density function if

- ① $f(x) \geq 0$
- ② $\int_{-\infty}^{\infty} f(x) dx = 1$
- ③ $f(x)$ is piecewise continuous

- Given a density function f , there is a continuous random variable X with PDF as f
- Support of the random variable X with PDF f_X is $\text{supp}(X) = \{x : f_X(x) > 0\}$

$\int f_X(x) dx = 1$

- ▶ $\text{supp}(X)$ contains intervals in which X can fall with positive probability
- ▶ Remember: $P(X = x) = 0$ for a continuous random variable
- For a random variable X with PDF f_X , an event A is a subset of the real line and its probability is computed as $P(A) = \int_A f(x) dx$

Problem

Consider the function

$$f(x) = \begin{cases} 3x^2 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

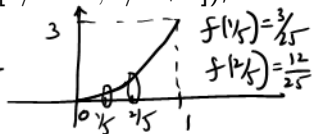
$$f(x) \geq 0$$

$$\int_0^1 3x^2 dx = x^3 \Big|_0^1 = 1$$

Show that f is a density function. Consider a random variable X with density f . Find $P(X = 1/5)$, $P(X = 2/5)$, $P(X \in [1/5 - \epsilon, 1/5 + \epsilon])$, $P(X \in [2/5 - \epsilon, 2/5 + \epsilon])$.

$$P(X = 1/5) = 0, P(X = 2/5) = 0$$

$$P\left(\frac{1}{5} - \epsilon < X < \frac{1}{5} + \epsilon\right) = \int_{\frac{1}{5} - \epsilon}^{\frac{1}{5} + \epsilon} 3x^2 dx = x^3 \Big|_{\frac{1}{5} - \epsilon}^{\frac{1}{5} + \epsilon}$$



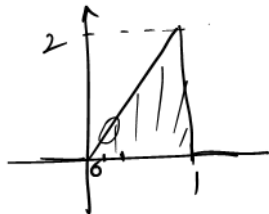
$$= \left(\frac{1}{5} + \epsilon\right)^3 - \left(\frac{1}{5} - \epsilon\right)^3 = \frac{6}{25}\epsilon + 2\epsilon^3 < \epsilon$$

$$P\left(\frac{2}{5} - \epsilon < X < \frac{2}{5} + \epsilon\right) = \left(\frac{2}{5} + \epsilon\right)^3 - \left(\frac{2}{5} - \epsilon\right)^3 = \frac{24}{25}\epsilon + 2\epsilon^3 < \epsilon$$

Problem

Consider a random variable X with density

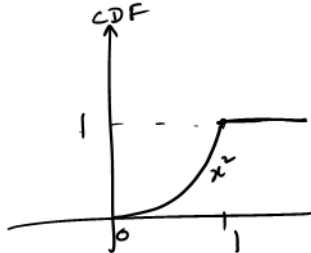
$$f_X(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$



Find $P(X \in [0.1, 0.3])$, $P(X \in (0.1, 0.03])$, $P(X \in [0.1, 0.03))$, $P(X \in (0.1, 0.03))$.

$$P(0.1 \leq X \leq 0.3) = \int_{0.1}^{0.3} 2x dx = x^2 \Big|_{0.1}^{0.3} = 0.09 - 0.01 = 0.08$$

$$P(0.8 \leq X \leq 1) = 1^2 - 0.8^2 = 0.36$$

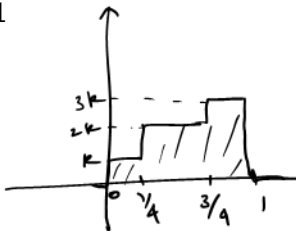


Problem

Consider the function

$$f(x) = \begin{cases} k & 0 \leq x < 1/4 \\ 2k & 1/4 \leq x < 3/4 \\ 3k & 3/4 \leq x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find k such that $f(x)$ is a valid density function.



$$\begin{aligned} \int_0^1 f(x) dx &= 1 \\ \int_0^{1/4} k dx + \int_{1/4}^{3/4} 2k dx + \int_{3/4}^1 3k dx &= 1 \\ \frac{1}{4}k + 2k \cdot \frac{1}{2} + 3k \cdot \frac{1}{4} &= 1 \end{aligned}$$

$$k = \frac{1}{2}$$

Common distributions: Uniform

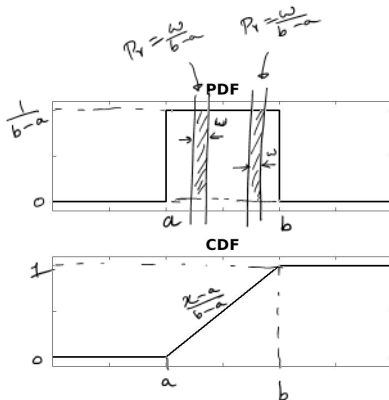
$$X \sim \text{Uniform}[a, b]$$

- PDF

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}$$

- CDF

$$F_X(x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x < b \\ 1 & x \geq b \end{cases}$$



Probability computations with uniform distribution

Suppose $X \sim \text{Uniform}[-10, 10]$. Find $P(-3 \leq X \leq 2)$, $P(5 < |X| < 7)$, $P(1 - \epsilon < X < 1 + \epsilon)$, $P(9 - \epsilon < X < 9 + \epsilon)$, $P(X > 7 | X > 3)$.

$$f_X(x) = \begin{cases} \frac{1}{20} & , -10 \leq x \leq 10 \\ 0 & , \text{otherwise} \end{cases}$$

$$P(-3 \leq X \leq 2) = \frac{5}{20} \quad , \quad P(5 < |X| < 7) = P(5 < X < 7) + P(-7 < X < -5) \\ = \frac{2}{20} + \frac{2}{20} = \frac{4}{20}$$

$$P(x_0 - \epsilon < X < x_0 + \epsilon) = \frac{2\epsilon}{20}$$

x_0 inside $[-9, 9]$
 ϵ small

$$P(X > 7 | X > 3) = \frac{P(X > 7 \text{ and } X > 3)}{P(X > 3)} = \frac{P(X > 7)}{P(X > 3)} = \frac{3/20}{7/20} = \frac{3}{7}$$

$\downarrow$
 3×4 7×4

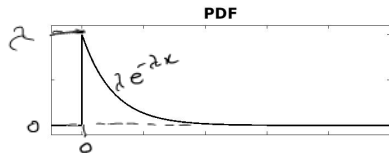
$$P(X > 4 | X > 0) = \frac{P(X > 4)}{P(X > 0)} = \frac{6/20}{10/20} = \frac{3}{5}$$

Common distributions: Exponential

$$X \sim \text{Exp}(\lambda)$$

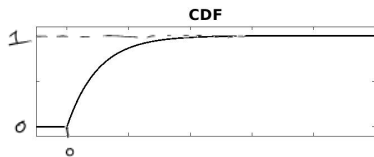
- PDF

$$f_X(x) = \begin{cases} \lambda \exp(-\lambda x) & x > 0 \\ 0 & \text{otherwise} \end{cases}$$



- CDF

$$F_X(x) = \begin{cases} 0 & x \leq 0 \\ 1 - \exp(-\lambda x) & x > 0 \end{cases}$$



$$\begin{aligned} F_X(x_0) &= \int_0^{x_0} \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_0^{x_0} \\ &= 1 - e^{-\lambda x_0} \end{aligned}$$

Probability computations with exponential distribution

Suppose $X \sim \text{Exp}(2)$. Find $P(5 < X < 7)$, $P(1 - \epsilon < X < 1 + \epsilon)$, $P(9 - \epsilon < X < 9 + \epsilon)$, $P(X > 4)$, $P(X > 7 | X > 3)$.

$$f_X(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad P(5 < X < 7) = \int_5^7 2e^{-2x} dx = -e^{-2x} \Big|_5^7 = e^{-10} - e^{-14}$$

$e \approx 2.7 \dots$

$$P(1 - \epsilon < X < 1 + \epsilon) = e^{-2(1-\epsilon)} - e^{-2(1+\epsilon)} = e^{-2} (e^{2\epsilon} - e^{-2\epsilon})$$

$$P(9 - \epsilon < X < 9 + \epsilon) = e^{-2(9-\epsilon)} - e^{-2(9+\epsilon)} = e^{-18} (e^{2\epsilon} - e^{-2\epsilon}) \ll P(1 - \epsilon < X < 1 + \epsilon)$$

$$P(X > 4) = 1 - P(X \leq 4) = 1 - (1 - e^{-8}) = e^{-8}$$

$$P(X > 7 | X > 3) = \frac{P(X > 7)}{P(X > 3)} = \frac{e^{-14}}{e^{-6}} = e^{-8}$$

$$P(X > s+t | X > s) = e^{-t}$$

Unique to exponential distribution
"memoryless"
independent of 's'

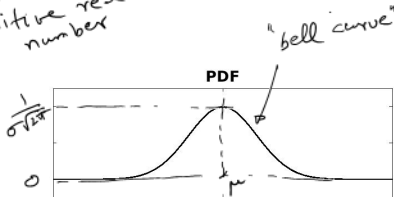
Common distributions: Normal (or) Gaussian

$$X \sim \text{Normal}(\mu, \sigma^2)$$

μ : any real number
 σ : positive real number

- PDF $\cdot \text{Supp}(X) = \mathbb{R}$

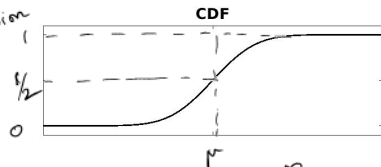
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



- CDF

$$F_X(x) = \int_{-\infty}^x f_X(u) du$$

No closed form expression



- Standard normal: $\text{Normal}(0, 1)$

μ σ

Exercise: Check $\int_{-\infty}^{\infty} f_X(x) dx = 1$
 $P(X < \mu) = P(X > \mu) = 1/2$

Probability computations with normal distribution

- CDF of $X \sim \text{Normal}(\mu, \sigma^2)$ does not have a closed form expression

- Standardization: If $X \sim \text{Normal}(\mu, \sigma^2)$, then

$$Z = (X - \mu)/\sigma \sim \text{Normal}(0, 1)$$

- ▶ $Z \sim \text{Normal}(0, 1)$, PDF: $f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp(-z^2/2)$

$$\text{CDF: } F_Z(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp(-u^2/2) du$$

- ▶ Normal table: Tabulation of above function $F_Z(z)$

★ Available on most computing systems

- How to compute probabilities for a normal distribution?

- ▶ Convert probability computation to that of a standard normal
- ▶ Use normal tables or computing systems

Problem

$$\mu=2, \sigma^2=5$$

Suppose $X \sim \text{Normal}(2, 5)$. Find $P(X < 5)$, $P(X < 10)$, $P(X < -5)$, $P(X < -10)$, $P(X > 5)$, $P(X > 10)$.

$$Z = \frac{X-2}{\sqrt{5}} \sim \text{Normal}(0, 1)$$

$$X = \sqrt{5} Z + 2$$

Assume: $F_Z(z)$ (CDF of Z) is known

$$X < 5 \Leftrightarrow \sqrt{5} Z + 2 < 5 \Leftrightarrow Z < 3/\sqrt{5}$$
$$P(X < 5) = P(Z < 3/\sqrt{5}) = F_Z(3/\sqrt{5})$$

$$X > 5 \Leftrightarrow \sqrt{5} Z + 2 > 5 \Leftrightarrow Z > 3/\sqrt{5}$$
$$P(X > 5) = P(Z > 3/\sqrt{5}) = 1 - F_Z(3/\sqrt{5})$$

Problem

$$\mu=3, \sigma=1$$

Suppose $X \sim \text{Normal}(3, 1)$. Find $P(5 < X < 7)$, $P(-5 < X < 5)$, $P(1 - \epsilon < X < 1 + \epsilon)$, $P(9 - \epsilon < X < 9 + \epsilon)$, $P(X > 4)$, $P(X > 7 | X > 3)$.

$$Z = X - 3 \sim \text{Normal}(0, 1) \quad X = Z + 3$$

$$5 < X < 7 \Leftrightarrow 5 < Z + 3 < 7 \Leftrightarrow 2 < Z < 4$$

$$P(5 < X < 7) = P(2 < Z < 4) = F_Z(4) - F_Z(2)$$

$$\begin{aligned} P(X > 7 | X > 3) &= \frac{P(X > 7)}{P(X > 3)} = \frac{P(Z > 4)}{P(Z > 0)} = \frac{1 - F_Z(4)}{1/2} \\ &= 2(1 - F_Z(4)) \end{aligned}$$